

Day – 4 Network Threats and Mitigation

Task – 1 Listing the network treats.

- Here are 5 common network threats along with real-world examples from popular apps and services, formatted as a clear list:
- 1. Phishing
 - Description: Disguising as a trusted entity to trick users into revealing sensitive information like passwords or credentials.
 - Real-World Example (Instagram): Attackers sending fake copyright infringement or verification badge emails that lead to lookalike login portals to harvest user credentials.
- 2. DDoS (Distributed Denial of Service)
 - Description: Overwhelming a server or network with a massive flood of traffic to render the service unavailable to legitimate users.
 - Real-World Example (Zomato): A botnet flooding food-ordering APIs with millions of simultaneous automated requests during a major festival sale or peak dinner hour to crash checkout systems.
- 3. Man-in-the-Middle (MitM)
 - Description: Secretly intercepting and relaying communications between two parties who believe they are directly communicating with each other.
 - Real-World Example (WhatsApp): An attacker on an unsecured public Wi-Fi network intercepting unencrypted metadata or exploiting network routing gaps to eavesdrop on data packets.
- 4. Malware
 - Description: Malicious software (such as trojans, spyware, or keyloggers) designed to infiltrate systems and steal data.
 - Real-World Example (Instagram): Infostealer malware downloaded via pirated PC software that silently scrapes saved browser cookies and session tokens to hijack accounts without triggering password alerts.
- 5. Password Attacks
 - Description: Attempts to crack or compromise user passwords using brute-force, dictionary lists, or credential stuffing.
 - Real-World Example (Paytm): Attackers using automated scripts to test lists of leaked username and password combinations (from unrelated data breaches) to break into user wallets.

Task – 2 Brute-force attack scenario.

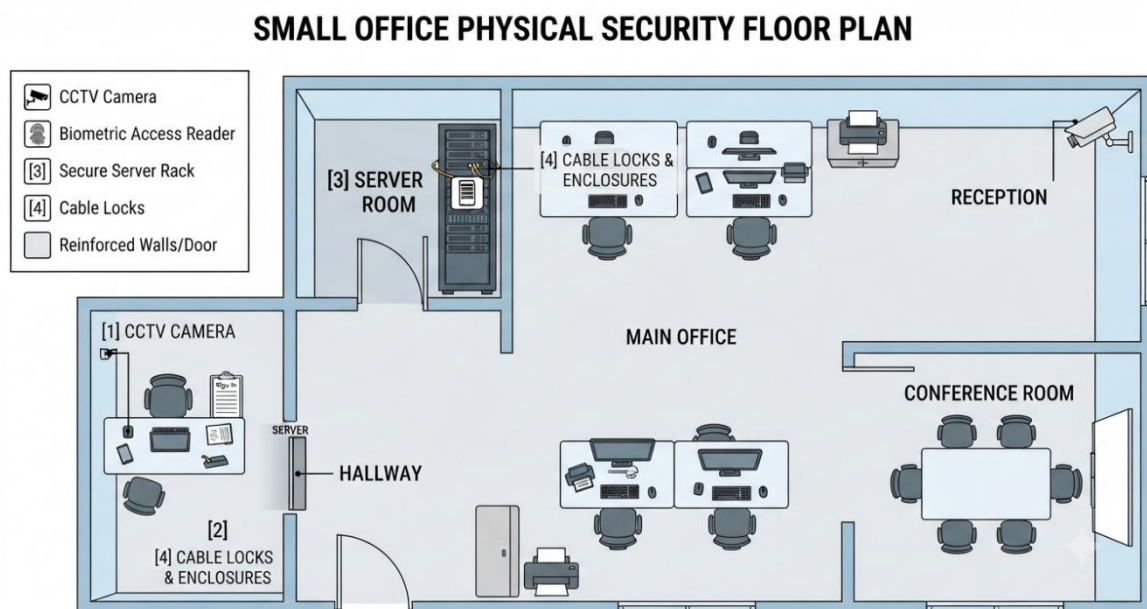
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- def brute_force_pin(target_pin):
-     """
-     Simulates a brute-force attack on a 4-digit PIN.
-     """
-     attempts = 0
-     print(f"Target PIN set to: {target_pin}")
-     print("Initiating brute-force attack...\n")
-     # Loop through all numbers from 0 to 9999
-     for i in range(10000):
-         # Format the integer as a 4-digit string with leading zeros (e.g., 7 -> "0007")
-         guess = f"{i:04d}"
-         attempts += 1
-         # Check if the guess matches the target PIN
-         if guess == target_pin:
-             print(f"[SUCCESS] PIN guessed correctly: {guess}")
-             print(f"Total attempts required: {attempts}")
-             return attempts
-     print("[FAILURE] PIN not found.")
-     return None
- # --- Example Execution ---
- if __name__ == "__main__":
-     # Preset secret PIN for the simulation
-     secret_pin = "4829"
-     brute_force_pin(secret_pin)
```

Task – 3 List 4 physical security risks.

- 1. Unauthorized Personnel Access
- The Risk: Insiders, disgruntled employees, or tailgating visitors gaining physical entry to the server racks to tamper with hardware, attach rogue keystroke loggers, or steal storage drives containing user data.
- Mitigation: Implement strict multi-factor access control systems (such as biometric scanners combined with RFID badge readers) and maintain continuous CCTV surveillance with automated tailgating detection.
- 2. Environmental Disasters (Fire and Overheating)
- The Risk: Electrical faults or failing cooling units causing localized fires or extreme temperature spikes, which can permanently destroy critical server hardware and cause massive service outages.

- Mitigation: Install a clean-agent gas fire suppression system (like Novec 1230 or FM-200) that extinguishes fires without leaving water residue, paired with redundant HVAC units and real-time environmental sensors.
- 3. Power Disruptions and Grid Failures
- The Risk: Sudden blackouts, voltage surges, or lightning strikes cutting power to the servers abruptly, leading to database corruption, unsaved transaction logs, and sudden downtime for users trying to book tickets or order food.
- Mitigation: Deploy an enterprise-grade Uninterruptible Power Supply (UPS) system backed by automatic generator failover mechanisms to ensure seamless power continuity during grid failures.
- 4. Water Damage and Leaks
- The Risk: Burst water pipes, leaking overhead air conditioning condensation lines, or external flooding compromising sensitive electronic server nodes and causing short circuits.
- Mitigation: Install raised flooring to keep critical hardware elevated off the ground, and run moisture-detection rope sensors underneath floors and around water-bearing pipes to alert administrators immediately upon leakage.

Task – 4 List security measures in small office.



- [1] CCTV Cameras: Positioned in the reception area and the main hallway to monitor activity and deter unauthorized access to critical areas.
- [2] Biometric Access: Installed on the door to the dedicated Server Room, ensuring that only authorized personnel can enter the space where core network equipment is housed.

- [3] Secure Server Rack & [4] Cable Locks: Within the Server Room, the server rack itself is locked, and individual equipment is secured with cable locks and enclosures to prevent tampering or theft of specific devices.

Task – 5 Mitigate risks from both network treats.

- Here is a tailored 7-point security checklist designed specifically for a college computer lab environment, balancing physical hardware protection with robust network defense against student misuse and external cyber threats:
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- 1. Enforce Workstation Port Security and Hardware Tethering
- What to do: Bolt computer towers and monitors directly to desks using secure steel cables and locking brackets, and use physical port blockers or software controls to disable unused USB ports.
- Why it matters: Prevents students or intruders from physically stealing peripherals or plugging in unauthorized external drives loaded with keyloggers or malware.
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- 2. Strip Local Administrator Privileges
- What to do: Configure all student-facing lab terminals to run under standard, restricted user accounts while reserving administrator rights strictly for IT staff.
- Why it matters: Stops users from installing unauthorized software, bypassing security controls, or altering system configurations that could expose the local network to vulnerabilities.
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- 3. Implement Deep-Freeze or Automated Re-imaging Software
- What to do: Deploy snapshot-based recovery software (such as Deep Freeze) or automated network imaging that wipes local changes and restores the operating system to a clean baseline every time a PC reboots.
- Why it matters: Instantly neutralizes any persistent malware, downloaded spyware, or accidental file corruption left behind by users during a lab session.
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- 4. Deploy Network Access Control (NAC) and Port Authentication
- What to do: Implement IEEE 802.1X authentication and MAC address filtering on all lab network switches to ensure only authorized, domain-joined lab machines can access the internet and internal campus resources.
- Why it matters: Prevents students from plugging their personal laptops or unauthorized rogue routers directly into the lab's ethernet wall jacks to bypass campus firewalls or sniff network traffic.
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- 5. Secure the Lab's Infrastructure Rack
- What to do: House local network switches, patch panels, and edge routers inside a locked wire mesh or solid steel server rack located within the lab supervisor's sightline or a secure utility closet.
- Why it matters: Protects critical network hardware from being physically disconnected, tampered with, or vandalized by disgruntled individuals.
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- 6. Restrict Lab Access via Biometric or RFID Door Readers
- What to do: Install electronic card or biometric scanners at the computer lab entrance, coupled with automated door closures, to restrict entry strictly to enrolled students and authorized faculty during scheduled hours.
- Why it matters: Mitigates tailgating and unauthorized after-hours physical entry, protecting both the expensive equipment inside and students' personal belongings.
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- 7. Monitor Network Traffic and Session Logs
- What to do: Set up centralized logging and intrusion detection systems (IDS) to monitor bandwidth consumption, flag unusual outbound traffic patterns, and track authentication attempts across lab terminals.
- Why it matters: Helps network administrators quickly identify compromised accounts, unauthorized data scraping, or distributed attacks originating from within the lab environment.
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